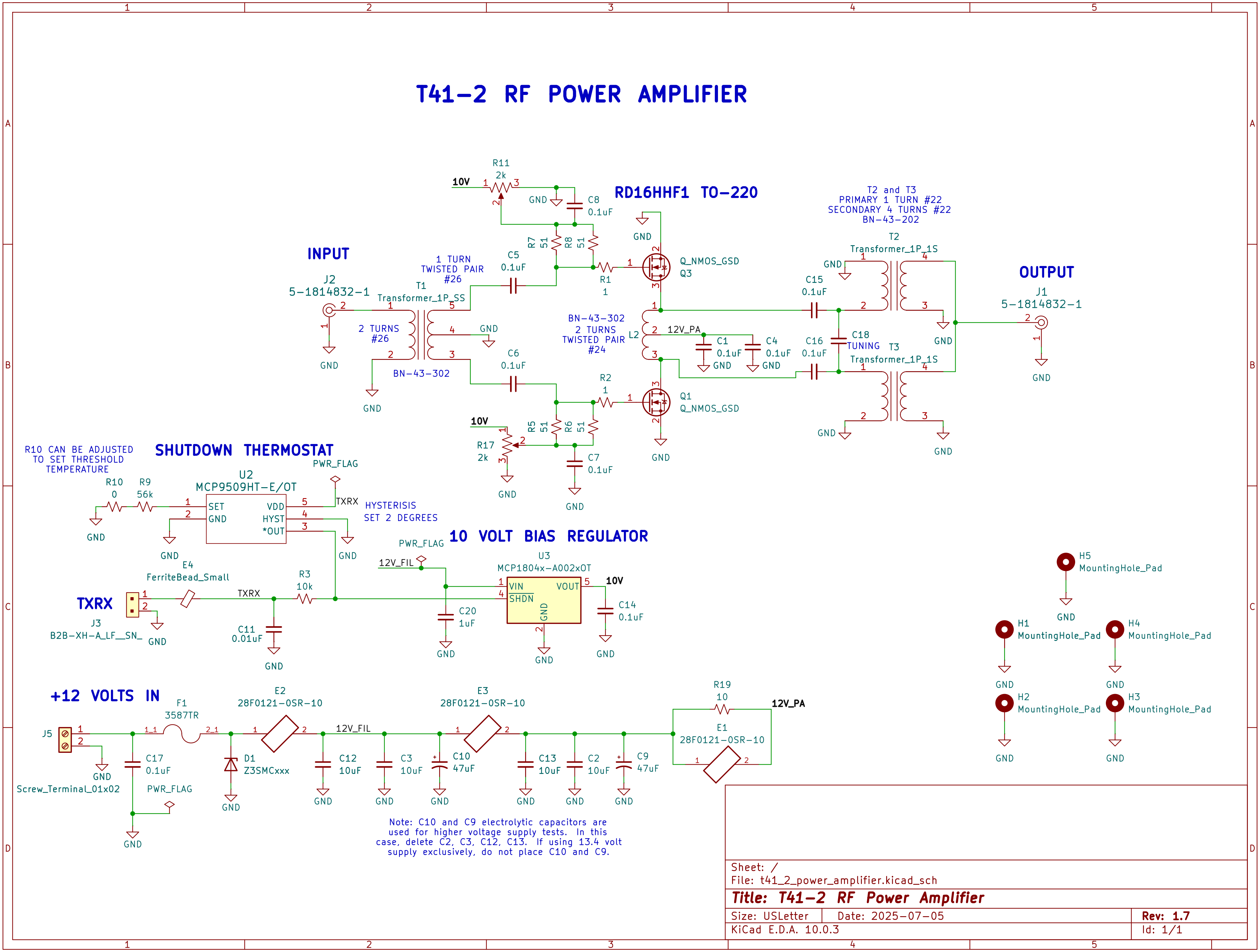
T41-2 RF POWER AMPLIFIER
INPUT
J2
5-1814832-1
1 TURN
TWISTED PAIR
#26
T1
Transformer_1P_SS
2 TURNS
#26
BN-43-302
C5
0.1uF
C6
0.1uF
R11
2k
10V
GND
C8
0.1uF
R7
51
R8
51
R1
1
RD16HHF1 TO-220
Q_NMOS_GSD
Q3
BN-43-302
2 TURNS
TWISTED PAIR
#24
L2
12V_PA
C1
0.1uF
C4
0.1uF
R2
1
Q1
Q_NMOS_GSD
R17
2k
R5
51
R6
51
C7
0.1uF
T2 and T3
PRIMARY 1 TURN #22
SECONDARY 4 TURNS #22
BN-43-202
T2
Transformer_1P_1S
C15
0.1uF
C16
0.1uF
C18
TUNING
T3
Transformer_1P_1S
OUTPUT
J1
5-1814832-1
R10 CAN BE ADJUSTED
TO SET THRESHOLD
TEMPERATURE
SHUTDOWN THERMOSTAT
U2
MCP9509HT-E/OT
R10
0
R9
56k
SET
GND
VDD
HYST
*OUT
PWR_FLAG
TXRX
HYSTERISIS
SET 2 DEGREES
10 VOLT BIAS REGULATOR
U3
MCP1804x-A002xOT
12V_FIL
VIN
SHDN
VOUT
C20
1uF
C14
0.1uF
E4
FerriteBead_Small
R3
10k
TXRX
J3
B2B-XH-A_LF__SN_
C11
0.01uF
H5
MountingHole_Pad
H1
MountingHole_Pad
H4
MountingHole_Pad
H2
MountingHole_Pad
H3
MountingHole_Pad
+12 VOLTS IN
J5
Screw_Terminal_01x02
F1
3587TR
C17
0.1uF
D1
Z3SMCxxx
E2
28F0121-0SR-10
E3
28F0121-0SR-10
C12
10uF
C3
10uF
C10
47uF
C13
10uF
C2
10uF
C9
47uF
R19
10
E1
28F0121-0SR-10
Note: C10 and C9 electrolytic capacitors are
used for higher voltage supply tests. In this
case, delete C2, C3, C12, C13. If using 13.4 volt
supply exclusively, do not place C10 and C9.
Sheet: /
File: t41_2_power_amplifier.kicad_sch
Title: T41-2 RF Power Amplifier
Size: USLetter
Date: 2025-07-05
KiCad E.D.A. 10.0.3
Rev: 1.7
Id: 1/1